



Regulatory Complaint Investigation and CAPA Report

Confidential - AI Draft

Document No.	DP-M4-20260708-DEVICEBR
Classification	Confidential - AI Draft
Generated	2026-07-08T07:42:52Z
Device	DocPlus+ Pulse Oximeter (DEV-PULSE-OX)
Firmware	v3.4
Framework	ISO 13485
Prepared By	DocPlus+ M4 Evidence and Documentation Layer

This document was generated from live M1 graph facts and M2 agent outputs. Every summarized claim is tied to captured agent state, evidence citations, or audit findings.

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This output follows a formal regulatory-document pattern: controlled metadata first, then complaint facts, investigation logic, objective evidence, risk decision, CAPA plan, review gates, and approval placeholders.

1. Document Profile and Controlled Metadata

Document Title	DocPlus+ Regulatory Complaint Investigation and CAPA Report
Document Number	DP-M4-20260708-DEVICEBR
Revision	A - AI Generated Draft
Complaint ID	CPL-20260708-3894
CAPA ID	CAPA-20260708-4268
RTM ID	RTM-DP-38944268
Classification	Confidential - AI Draft for quality and regulatory review
Generated Timestamp	2026-07-08T07:42:52Z
System of Record	DocPlus+ M4 Evidence and Documentation Layer
Device	DocPlus+ Pulse Oximeter (DEV-PULSE-OX)
Current Firmware	v3.4
Affected Component	COMP-MEASUREMENT
Regulatory Framework	ISO 13485

Data Quality and Source Provenance

This report references 46 graph fact(s): 15 controlled, 25 extracted, 4 inferred, 2 internal.

Uncontrolled graph facts appear in this report and are visibly tagged. [DEMO DATA - NOT CONTROLLED EVIDENCE] means the item cannot support CAPA closure without controlled verification. [INFERRED DATA - REVIEW REQUIRED] means the item was inferred and requires reviewer confirmation before final use.

Source Type	Referenced Fact Count
controlled	15
extracted	25
inferred	4
synthetic	0
internal	2

Digital-Twin Prototype Disclaimer

Hackathon prototype disclaimer: this AI-generated investigation may include digital-twin or firmware-traceability simulation, not real hardware verification. Simulated results may support investigation triage, but they require formal human engineering approval before any real-world quality, regulatory, release, or closure reliance.

Document Purpose

This report documents the complaint intake, AI-assisted investigation record, retrieved objective evidence, risk decision, CAPA recommendations, audit-style gaps, and corrective action controls. It is intended as a controlled draft that a Quality or Regulatory reviewer can verify, approve, or return for correction.

2. Regulatory Format Controls

ALCOA+ Data Integrity Mapping

Principle	DocPlus+ Implementation
Attributable	Prepared by DocPlus+ M4; generated timestamp and device scope are recorded.
Legible	A4 PDF with selectable text, structured headings, repeatable tables, and page numbers.
Contemporaneous	Generated at the time of complaint-report request from live agent state.
Original	Marked as AI Draft until Quality and Regulatory reviewers approve.
Accurate	Risk, CAPA, evidence, audit, and trace sections are populated from structured state.
Complete	Includes intake, investigation, evidence, risk, CAPA, audit, trace, and approval fields.
Consistent	Uses the same DocPlus+ controlled layout for all complaint-report exports.
Enduring	Saved as a PDF artifact under the application output directory.
Available	Downloadable through the M4 document endpoint and traceable by document number.

Applicable Regulatory Basis

Framework	Relevance to This Report
ISO 13485	Complaint handling, CAPA discipline, documented information, traceable quality records.
21 CFR Part 820	Design controls, complaint files, CAPA, and objective evidence expectations.
EU MDR	Technical documentation, post-market surveillance, vigilance, and risk-benefit continuity.
21 CFR Part 11	Electronic record controls, auditability, attribution, and durable retrieval.

3. Executive Summary

Severity Low	Risk Medium / RPN 10	Reportable False
Evidence Confidence medium (score 0.86)	Evidence Class Mix Candidate evidence - requires Quality review: 1, Controlled current verification: 11, Historical controlled evidence - stale or prior version: 1, Firmware traceability signal - informational, not controlled verification evidence: 2	Evidence Items 15
Audit Findings 1	Trace Decay Alerts 1	

Complaint: Device brightness cant be adjusted

Agent conclusion: Complaint: DEV-PULSE-OX brightness cannot be adjusted (severity: Low). Most probable cause is a software configuration/write-path issue similar to prior incident CPL-100-013, where brightness menu writes were rejected after a configuration CRC mismatch in the display preference block. No unit-specific objective evidence (serial, firmware, logs, return evaluation) is present to confirm this for the current case. Hardware and firmware-compatibility hypotheses remain lower probability based on the available facts.

Risk Evidence Confidence Basis

Weighted evidence-class score 0.86 from 15 item(s): Candidate evidence - requires Quality review=1, Controlled current verification=11, Historical controlled evidence - stale or prior version=1, Firmware traceability signal - informational, not controlled verification evidence=2. CAPA-blocking evidence-class item count: 4. Rule: Risk evidence confidence is the average of per-item evidence-class weights (controlled_verification=1.00, historical_controlled=0.70, simulated=0.45, candidate=0.25, no_evidence=0.00). The final label is capped below high when any CAPA-closure-blocking evidence class is present, because those items cannot support final closure until controlled verification is attached.

Confidence drivers: RUN-TC-MEAS-022-V3-4: controlled_verification, RUN-TC-MEAS-030-V3-4: controlled_verification, RUN-TC-MEAS-032-V3-4-2: historical_controlled, RUN-TC-MEAS-021-V3-4: controlled_verification, RUN-TC-MEAS-025-V3-4: controlled_verification, RUN-TC-MEAS-029-V3-4: controlled_verification, RUN-TC-MEAS-034-V3-4: controlled_verification, RUN-TC-MEAS-026-V3-4: contro...

Quality Review Decision

The complaint should remain open until objective evidence is reconciled, CAPA ownership is assigned, and any major or critical audit findings affecting the complaint scope are remediated or justified.

4. Complaint Investigation Record

Field	Generated Value	Regulatory Basis
Complaint ID	CPL-20260708-3894	ISO 13485 8.2.2
Date Received	2026-07-08T07:42:52Z	21 CFR Part 11
Device UDI	UDI-DI-08717648-207970	EU MDR Article 27 / 21 CFR 830
Model / Version	DOCPLUS+-PULSE-OXIMETER / v3.4	IEC 62304 Section 8
Complaint Source	Post-market complaint intake	ISO 13485 8.2.2
Complaint Narrative	Device brightness cant be adjusted	21 CFR 820 complaint file
Patient Impact	Potential device performance or usability impact requiring quality review.	EU MDR Article 87
MDR Reportable?	No	EU MDR Article 87 / 21 CFR 803
CAPA Initiated?	Yes - CAPA-20260708-4268	21 CFR 820.100
Affected Component	COMP-MEASUREMENT	ISO 13485 8.5.2
Closure Status	Pending - Insufficient Evidence: Closure blocked: evidence class requires controlled verification before closure: RUN-TC-MEAS-032-V3-4-2, CAPA-CPL-100-013.	ISO 13485 8.2.2

Reportability Decision Logic

Condition	DocPlus+ Assessment	Action
Death or serious injury	Not stated in complaint text.	Monitor for escalation during intake and investigation.
Potential harm or near miss	Potential device performance or usability impact requiring quality review.	Document rationale and continue risk review.
Device relationship	Complaint maps to System Reliability.	Keep CAPA open until engineering evidence confirms or rules out relationship.
Malfunction/use issue	Reported condition indicates possible device malfunction or use-related failure.	Verify recurrence risk and effectiveness of controls.

Related Prior Complaints

Complaint ID	Similarity	Summary	Source
CPL-100-013	4	Display brightness cannot be adjusted. [INFERRED DATA - REVIEW REQUIRED]	inferred [INFERRED DATA - REVIEW REQUIRED]
CPL-100-035	2	Readings differ significantly from clinical device. [INFERRED DATA - REVIEW REQUIRED]	inferred [INFERRED DATA - REVIEW REQUIRED]
CPL-100-056	2	Alarm volume cannot be adjusted. [INFERRED DATA - REVIEW REQUIRED]	inferred [INFERRED DATA - REVIEW REQUIRED]

5. Requirements Traceability Matrix

Req ID	Requirement	Scope / Source	Relevance	Evidence	Test Case	Status / Result
REQ-MEA S-035	Readings differ significantly from clinical device.	strong - RTM_M aster.xlsx	Matched complaint terms against measurement_core; relevance score 8.	verified_current - Controlled current verification	TC-MEAS-035	current and usable - Pass
REQ-MEA S-026	Device continuously searches for signal.	strong - RTM_M aster.xlsx	Matched complaint terms against measurement_core; relevance score 6.	verified_current - Controlled current verification	TC-MEAS-026	current and usable - Pass
REQ-MEA S-029	Device reports impossible SpO2 values.	strong - RTM_M aster.xlsx	Matched complaint terms against measurement_core; relevance score 6.	verified_current - Controlled current verification	TC-MEAS-029	current and usable - Pass
REQ-MEA S-034	Device cannot detect weak pulse.	strong - RTM_M aster.xlsx	Matched complaint terms against measurement_core; relevance score 6.	verified_current - Controlled current verification	TC-MEAS-034	current and usable - Pass
REQ-MEA S-032	Perfusion index is missing.	strong - RTM_M aster.xlsx	Matched complaint terms against measurement_core; relevance score 5.	stale - Historical controlled evidence - stale or prior version	TC-MEAS-032	stale - Pass
REQ-DISP -013	Display brightness cannot be adjusted.	medium - RTM_Master.xlsx	Matched complaint terms against display_task; relevance score 4.	verified_current - Controlled current verification	TC-DISP-013	current and usable - Pass
REQ-MEA S-021	SpO2 reading remains at 0%.	medium - RTM_Master.xlsx	Matched complaint terms against measurement_core; relevance score 4.	verified_current - Controlled current verification	TC-MEAS-021	current and usable - Pass
REQ-MEA S-022	Pulse rate is not detected.	medium - RTM_Master.xlsx	Matched complaint terms against measurement_core; relevance score 4.	verified_current - Controlled current verification	TC-MEAS-022	current and usable - Pass
REQ-MEA S-023	SpO2 reading fluctuates excessively.	medium - RTM_Master.xlsx	Matched complaint terms against measurement_core; relevance score 4.	verified_current - Controlled current verification	TC-MEAS-023	current and usable - Pass
REQ-MEA S-024	Pulse rate reading jumps randomly.	medium - RTM_Master.xlsx	Matched complaint terms against measurement_core; relevance score 4.	verified_current - Controlled current verification	TC-MEAS-024	current and usable - Pass
REQ-MEA S-025	Measurement takes too long to stabilize.	medium - RTM_Master.xlsx	Matched complaint terms against measurement_core; relevance score 4.	verified_current - Controlled current verification	TC-MEAS-025	current and usable - Pass
REQ-MEA S-027	Finger detection fails.	medium - RTM_Master.xlsx	Matched complaint terms against measurement_core; relevance score 4.	verified_current - Controlled current verification	TC-MEAS-027	current and usable - Pass

Trace Decay Risk Statement

Rows marked STALE are not filler rows: they are generated from the live Trace Decay result and indicate verification links that require review before the device can claim current-firmware evidence.

6. Design History File Index

#	Element	Document ID	Generated Unique Content	Status
1	Planning	DDP-DP-38944268	Design plan for complaint-impact review	Open
2	Design Input	URS-DP-38944268	Complaint-linked requirement review for Device brightness cant be adjusted	Draft
3	Design Output	SDD-DP-38944268	System Reliability design output references mapped	Draft
4	Design Review	DRM-DP-38944268	Cross-functional review required	Pending
5	Verification	VPR-DP-38944268	Objective evidence required for System Reliability on v3.4	Open
6	Validation	VAL-DP-38944268	Use-condition validation required for Device brightness cant be adjusted	Open
7	Design Transfer	DTR-DP-38944268	Release readiness blocked until evidence reconciled	Pending
8	Design Changes	DCR-DP-38944268	Change/service impact assessment linked to System Reliability	Open
9	DHF Index	DHFI-DP-38944268	DocPlus+ index for DocPlus+ Pulse Oximeter	Generated

Validation Gap Check

The validation section remains open because the complaint describes a user-reported condition involving System Reliability. DocPlus+ therefore does not mark the DHF complete until validation, verification, or justified non-reproducibility evidence is linked.

7. EU MDR Technical Documentation - Annex II / III

Annex Ref	Section	DocPlus+ Generated Content
Annex II Section 1	Device Description	UDI UDI-DI-08717648-207970 scoped to complaint involving System Reliability.
Annex II Section 3	Design and Manufacturing Information	Design/change impact for System Reliability is carried into design output review.
Annex II Section 4	GSPR Compliance	Reported condition maps to safety/performance review: Device brightness cant be adjusted.
Annex II Section 5	Benefit-Risk Analysis	Medium risk complaint requires residual-risk review before closure.
Annex II Section 6.1	Verification and Validation	Objective verification evidence must be reconciled against the complaint scope.
Annex III Section 1.3	PMS / PSUR Input	Complaint is retained as a post-market signal for trend monitoring.
Annex III Section 1.5	Vigilance Reporting	Reportability remains open until human regulatory review confirms decision.

Team NB Structure Check

The package preserves a cohesive structure with section titles, metadata, and cross-references so that information is locatable by a reviewer rather than buried in a generic AI narrative.

Record Boundary

The report is generated only from live application state: M1 graph context, M2 agent outputs, M4 evidence retrieval, AuditShadow findings, and Trace Decay alerts. It does not certify regulatory closure until the signature block is completed by authorized reviewers.

8. Complaint Investigation Workflow and Execution Trace

8.1 Structured Complaint Intake Detail

Device ID	DEV-PULSE-OX
Affected Component	COMP-MEASUREMENT (COMP-MEASUREMENT)
Severity	Low
Timeline	not specified
Symptom Codes	SYMPTOM_DEVICE, SYMPTOM_BRIGHTNESS, SYMPTOM_CANT, SYMPTOM_ADJUSTED
Match Terms	cant, device
Similar Prior Complaints	3

Complaint Narrative

Device brightness cant be adjusted

Related Prior Complaints

Complaint ID	Similarity	Summary	Source
CPL-100-013	4	Display brightness cannot be adjusted. [INFERRED DATA - REVIEW REQUIRED]	inferred [INFERRED DATA - REVIEW REQUIRED]
CPL-100-035	2	Readings differ significantly from clinical device. [INFERRED DATA - REVIEW REQUIRED]	inferred [INFERRED DATA - REVIEW REQUIRED]
CPL-100-056	2	Alarm volume cannot be adjusted. [INFERRED DATA - REVIEW REQUIRED]	inferred [INFERRED DATA - REVIEW REQUIRED]

8.2 Investigation Chronology

Time	Agent	Action	Status	Detail
2026-07-08T07:35:49.059046Z	langgraph_supervisor	route to complaint_intake	info	Recorded
2026-07-08T07:35:49.060307Z	complaint_intake	extract structured complaint	started	Recorded
2026-07-08T07:38:44.175275Z	complaint_intake	extract structured complaint	completed	Elapsed 175115.5 ms
2026-07-08T07:38:44.175775Z	langgraph_supervisor	handoff check: similar incidents preserved for root cause	info	Structured event metadata recorded
2026-07-08T07:38:44.324631Z	langgraph_supervisor	route to root_cause	info	Recorded
2026-07-08T07:38:44.324773Z	root_cause	generate complaint-specific failure hypotheses	started	Recorded
2026-07-08T07:38:49.163896Z	root_cause	generate complaint-specific failure hypotheses	completed	Elapsed 4839.1 ms
2026-07-08T07:38:49.381252Z	langgraph_supervisor	route to firmware_traceability_ripple_check	info	Recorded
2026-07-08T07:38:49.381349Z	firmware_traceability_ripple_check	run firmware traceability ripple checks	started	Recorded
2026-07-08T07:38:49.382537Z	firmware_traceability_ripple_check	hypothesis capability check	info	Structured event metadata recorded
2026-07-08T07:39:00.488410Z	firmware_traceability_ripple_check	hypothesis capability check	info	Structured event metadata recorded
2026-07-08T07:39:00.489749Z	firmware_traceability_ripple_check	hypothesis capability check	info	Structured event metadata recorded
2026-07-08T07:39:00.490023Z	firmware_traceability_ripple_check	run firmware traceability ripple checks	completed	Elapsed 11108.7 ms
2026-07-08T07:39:00.830519Z	langgraph_supervisor	route to evidence	info	Recorded
2026-07-08T07:39:00.830783Z	evidence	retrieve and validate evidence	started	Recorded
2026-07-08T07:39:12.267081Z	evidence	component-scoped evidence retrieval	info	Structured event metadata recorded
2026-07-08T07:39:12.268163Z	evidence	retrieve and validate evidence	completed	Elapsed 11437.4 ms
2026-07-08T07:39:12.478050Z	langgraph_supervisor	route to risk	info	Recorded

Chronology Interpretation

The sequence confirms the complaint passed through intake, root-cause hypothesis generation, firmware traceability ripple checking, evidence retrieval, risk scoring, CAPA drafting, and final reasoning. This preserves a reviewer-readable process trail for the AI investigation rather than presenting the CAPA as an unsupported narrative.

Trace AI - Pipeline Observability

Trace AI is operational engineering telemetry for this report generation. It records pipeline execution, LLM metadata, and retrieval/tool metadata only. It is not clinical evidence, not regulatory objective evidence, and not the same as Trace Decay or the Firmware Traceability Ripple Check.

Total wall-clock time	420730.08 ms
Agent steps	10
LLM calls	4
Tool/retrieval calls	25
Tokens	22261 total (16532 prompt / 5729 completion)
Longest agent	complaint_intake (175115.31 ms)

Agent Step Trace

Step ID	Agent	Action	Status	Elapsed	Sub-events
step-fae817573a	complaint_intake	extract structured complaint	success	175115.31 ms	LLM 1; tools 4
step-f799166142	root_cause	generate complaint-specific failure hypotheses	success	4839.21 ms	LLM 1; tools 2
step-1da95a0d4a	firmware_traceability_ripple_...	run firmware traceability ripple checks	success	11108.73 ms	LLM 0; tools 2
step-5c155d0e7b	evidence	retrieve and validate evidence	success	11437.47 ms	LLM 0; tools 3
step-1daf5a84b1	risk	calculate ISO 14971 risk	success	1465.87 ms	LLM 0; tools 2
step-8b3f3ba1a0	capa	draft CAPA with citations	success	10438.73 ms	LLM 0; tools 4
step-78f1693803	openai_reasoning	final complaint reasoning	success	15354.84 ms	LLM 1; tools 0
step-30106fe4cf	audit_shadow	run adversarial audit	success	62282.47 ms	LLM 0; tools 5
step-5f32386e45	openai_reasoning	final audit reasoning	success	21607.15 ms	LLM 1; tools 0
step-438c6fd391	cybersecurity	scan SBOM components against NVD	success	107080.3 ms	LLM 0; tools 3

LLM Call Metadata

Call ID	Agent	Task	Model	Status	Tokens	Latency
llm-eabcc106b7	complaint_intake	symptom_classification	gpt-5	success	1608 (1312/296)	2864.49 ms
llm-4c7e25a188	root_cause	root_cause_hypotheses	gpt-4o-mini	success	5453 (4823/630)	4319.87 ms
llm-a2e50287d2	openai_reasoning	complaint_final_reasoning	gpt-5	success	7113 (4870/2243)	15346.65 ms
llm-a27e61d62f	openai_reasoning	audit_final_reasoning	gpt-5	success	8087 (5527/2560)	21605.56 ms

Tool and Retrieval Call Metadata

Call ID	Agent	Tool	Query / Target	Results	Latency
tool-02927bdc d0	complaint_intake	graph_tools.resolve_de vice_id	device_id=DEV-PULSE-OX		612.41 ms
tool-40d9565a b4	complaint_intake	graph_tools.resolve_de vice_id	device_id=DEV-PULSE-OX		678.2 ms
tool-3bd8716c 74	complaint_intake	graph_tools.get_device _context	device_id=DEV-PULSE-OX	15	167948.61 ms
tool-a6e5475d 83	complaint_intake	graph_tools.find_similar _complaints	term=Device brightness cant be adjusted SYMPTOM_DEVICE SYMPTOM_BRIGHTNE...	3	204.12 ms
tool-0984c2ea 7e	root_cause	graph_tools.get_graph_ neighborhood	node_id=COMP-MEASUREMENT	457	373.26 ms
tool-ee0e6d7c ec	root_cause	graph_tools.get_graph_ neighborhood	node_id=COMP-MEASUREMENT	34	138.68 ms
tool-27ae4f53 a2	firmware_traceabi lity_rip...	graph_tools.resolve_de vice_id	device_id=DEV-PULSE-OX		111.66 ms
tool-620d6fa1 ae	firmware_traceabi lity_rip...	graph_tools.run_ripple	device_id=DEV-PULSE-OX; new_version=v3.5; changed_components=['COMP-MEA...	21	11083.38 ms
tool-dfcd879cf e	evidence	graph_tools.resolve_de vice_id	device_id=DEV-PULSE-OX		296.66 ms
tool-6e26c6f3 80	evidence	graph_tools.get_compo nent_scoped_ev...	device_id=DEV-PULSE-OX; component_id=COMP-MEASUREMENT; component_name=C...	12	1836.07 ms
tool-556fbe09 41	evidence	vector_tools.retrieve_do cuments_debug	device_id=DEV-PULSE-OX; query=Device brightness cant be adjusted COMP-M...	8	9592.28 ms
tool-36d74f85 2a	risk	graph_tools.resolve_de vice_id	device_id=DEV-PULSE-OX		132.69 ms
tool-be07d2da 8a	risk	graph_tools.open_capa _count	device_id=DEV-PULSE-OX	2	1464.24 ms
tool-49f2f229f 1	capa	graph_tools.resolve_de vice_id	device_id=DEV-PULSE-OX		228.06 ms
tool-e3afc975 c1	capa	graph_tools.get_audit_s cope	device_id=DEV-PULSE-OX	120	2123.83 ms
tool-fccd20dd 60	capa	graph_tools.resolve_de vice_id	device_id=DEV-PULSE-OX		280.04 ms
tool-e9154d7b c6	capa	graph_tools.get_capa_c ontext	device_id=DEV-PULSE-OX	0	8289.49 ms
tool-7bbfd3b6 e4	audit_shadow	graph_tools.resolve_de vice_id	device_id=DEV-PULSE-OX		225.93 ms
tool-024c4a71 90	audit_shadow	graph_tools.resolve_de vice_id	device_id=DEV-PULSE-OX		204.53 ms
tool-7134e591 84	audit_shadow	graph_tools.get_device _context	device_id=DEV-PULSE-OX	15	41409.54 ms

5 additional tool/retrieval call(s) are retained in structured Trace AI data.

9. Root Cause Analysis

ID	Hypothesis	Probability	Affected Area	Key Why Chain
HYP-001	COMP-MEASUREMENT: Software Configuration Issue [INFERRED DATA - REVIEW REQUIRED]	60%	COMP-MEASUREME NT	Device attempts to communicate brightness change but receives no acknowledgment. Error in the brightness configuration may prevent changes from being registered. [INFERRED DATA - REVIEW REQUIRED]
HYP-002	COMP-MEASUREMENT: Brightness Adjustment Hardware Fault	30%	COMP-MEASUREME NT	Brightness control circuit may be faulty, preventing any adjustments from being made. Physical failure could lead to a lack of response from the adjustment commands.
HYP-003	COMP-MEASUREMENT: Firmware Compatibility Issue	10%	COMP-MEASUREME NT	Updates may introduce incompatibilities that affect brightness adjustments. If the interface layer of the firmware is not compatible, commands might fail.

HYP-001 Evidence Balance

Evidence For	Prior complaint CPL-100-013 indicates a similar issue with brightness adjustments related to configuration. [INFERRED DATA - REVIEW REQUIRED]
Evidence Against / Gaps	No evidence directly states that configuration settings are currently faulty for this specific component. [INFERRED DATA - REVIEW REQUIRED]
Similar Incident Cross-Check	CPL-100-013: Pattern is consistent, as the similar incident involved brightness configuration. [INFERRED DATA - REVIEW REQUIRED]
Probability Rationale	The connection to prior incidents increases probability, though no direct evidence confirms current fault. [INFERRED DATA - REVIEW REQUIRED]

HYP-002 Evidence Balance

Evidence For	No direct hardware evidence for the current incident, but similar physical faults have been observed in related incidents.
Evidence Against / Gaps	No physical inspection or return data referenced in the facts indicate a hardware failure.
Similar Incident Cross-Check	No direct related incidents, making the pattern insufficient to draw conclusions.
Probability Rationale	While hardware issues can occur, the exact nature of the current failure isn't sufficiently substantiated by evidence.

HYP-003 Evidence Balance

Evidence For	Prior complaint of firmware incompatibility with related component adjustments, such as CPL-100-056.
Evidence Against / Gaps	No specific evidence indicates firmware was the cause in this incident, nor is there clear pass/fail data from tests. Current software version comparisons do not indicate a mismatch in expected functionality.
Similar Incident Cross-Check	CPL-100-056: Inconsistent with brightness adjustment as it pertains to alarm volume instead.
Probability Rationale	Lower probability due to lack of direct evidence linking firmware to the specific complaint of brightness failure.

Root Cause Rationale

- Strong symptom match to CPL-100-013: identical user-facing failure ("Display brightness cannot be adjusted") with recorded root cause of configuration CRC mismatch blocking brightness preference writes.
- CPL-100-013 investigation notes indicate the corrected software path passed requirements-based and regression tests (e.g., TC-DISP-013) without reintroducing the failure, supporting a software/configuration mechanism as plausible for similar

reports.

3. No hardware-specific evidence for this complaint (no returned unit, no inspection findings) and no directly related brightness hardware incident in the similar set; the nearest hardware case (CPL-100-056) concerns alarm volume, not display brightness.
4. Firmware-compatibility hypothesis (HYP-003) is weakened by lack of version details for this unit and by unrelated similarity to CPL-100-056; furthermore, controlled verification evidence provided pertains to measurement features (e.g., RUN-TC-MEAS-021/025/029/030/032) and does not implicate brightness handling.
5. Available controlled verification runs passing on v3.4/v3.4.2 for measurement functions neither confirm nor refute brightness behavior; they simply do not provide evidence for a display-configuration failure, leaving HYP-001 as most plausible by pattern association rather than by direct test evidence for this unit.

Primary Root Cause Position

For this generated draft, the leading position is that the complaint is associated with System Reliability and the evidence/control set tied to the reported condition. The position remains provisional until engineering review confirms or rules out the suspected cause with objective evidence.

10. Evidence and Provenance

Source	Evidence Class	Support	Conf.	Excerpt	Citation
REQ-MEAS-022 / TC-MEAS-022 / RUN-TC-MEAS-022-V3-4	Controlled current verification	Yes	96%	REQ-MEAS-022: Pulse rate is not detected. Test TC-MEAS-022: Verification test TC-MEAS-022; result=Pass; firmware=v3.4. Evidence RUN-TC-MEAS-022-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regression FW-v3.4 Verif...	[Source: REQ-MEAS-022/TC-MEAS-022, Confidence: 0.96]
REQ-MEAS-030 / TC-MEAS-030 / RUN-TC-MEAS-030-V3-4	Controlled current verification	Yes	96%	REQ-MEAS-030: Pulse waveform is not displayed. Test TC-MEAS-030: Verification test TC-MEAS-030; result=Pass; firmware=v3.4. Evidence RUN-TC-MEAS-030-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regression FW-v3.4...	[Source: REQ-MEAS-030/TC-MEAS-030, Confidence: 0.96]
REQ-MEAS-032 / TC-MEAS-032 / RUN-TC-MEAS-032-V3-4-2	Historical controlled evidence - stale or prior version	Yes	96%	REQ-MEAS-032: Perfusion index is missing. Test TC-MEAS-032: Verification test TC-MEAS-032; result=Pass; firmware=v3.4.2. Evidence RUN-TC-MEAS-032-V3-4-2: result=Pass; firmware=v3.4.2. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regression FW-v3.4...	[Source: REQ-MEAS-032/TC-MEAS-032, Confidence: 0.96]
REQ-MEAS-021 / TC-MEAS-021 / RUN-TC-MEAS-021-V3-4	Controlled current verification	Yes	96%	REQ-MEAS-021: SpO2 reading remains at 0%. Test TC-MEAS-021: Verification test TC-MEAS-021; result=Pass; firmware=v3.4. Evidence RUN-TC-MEAS-021-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regression FW-v3.4 Verif...	[Source: REQ-MEAS-021/TC-MEAS-021, Confidence: 0.96]
REQ-MEAS-025 / TC-MEAS-025 / RUN-TC-MEAS-025-V3-4	Controlled current verification	Yes	96%	REQ-MEAS-025: Measurement takes too long to stabilize. Test TC-MEAS-025: Verification test TC-MEAS-025; result=Pass; firmware=v3.4. Evidence RUN-TC-MEAS-025-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regression...	[Source: REQ-MEAS-025/TC-MEAS-025, Confidence: 0.96]

Source	Evidence Class	Support	Conf.	Excerpt	Citation
REQ-MEAS-029 / TC-MEAS-029 / RUN-TC-MEAS-029-V3-4	Controlled current verification	Yes	96%	REQ-MEAS-029: Device reports impossible SpO2 values. Test TC-MEAS-029: Verification test TC-MEAS-029; result=Pass; firmware=v3.4. Evidence RUN-TC-MEAS-029-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regression FW...	[Source: REQ-MEAS-029/TC-MEAS-029, Confidence: 0.96]

10.1 Firmware Traceability Ripple Check

This section reports a firmware-traceability check: for each root-cause hypothesis, it examines whether linked verification test records match the complaint or candidate firmware version, or are stale/mismatched. This is a graph-based traceability signal, not a behavioral model or physical test of the device. It identifies where verification evidence may be out of date, not what the device would actually do under the reported failure condition.

Hypothesis	Analysis Type	Result	Interpretation	Evidence Class
HYP-001	Firmware traceability ripple check	fail	HYP-001 evaluates COMP-MEASUREMENT: Software Configuration Issue. The firmware traceability ripple check found 21 linked verification test record(s) that do not match candidate firmware v3.5; this flags stale or mismatched verification coverage for this hyp...	Firmware traceability signal - informational, not controlled verification evidence
HYP-002	Not applicable - measurement_calibration_or_physiology	not_run	HYP-002 evaluates COMP-MEASUREMENT: Brightness Adjustment Hardware Fault. The firmware traceability ripple check was not run because this failure mode maps to 'measurement_calibration_or_physiology', outside firmware-to-test traceability freshness review. T...	Firmware traceability signal - informational, not controlled verification evidence
HYP-003	Firmware traceability ripple check	fail	HYP-003 evaluates COMP-MEASUREMENT: Firmware Compatibility Issue. The firmware traceability ripple check found 21 linked verification test record(s) that do not match candidate firmware v3.5; this flags stale or mismatched verification coverage for this hyp...	Firmware traceability signal - informational, not controlled verification evidence

11. CAPA Report

Product	DocPlus+ Pulse Oximeter
Software Version	v3.4
CAPA Status	Pending - Insufficient Evidence
Priority	Low
Risk Classification	Moderate - complaint requires corrective action review
Evidence Confidence	medium (score 0.86)
Evidence Class Mix	Candidate evidence - requires Quality review: 1, Controlled current verification: 11, Historical controlled evidence - stale or prior version: 1, Firmware traceability signal - informational, not controlled verification evidence: 2

Risk Evidence Confidence Basis

Weighted evidence-class score 0.86 from 15 item(s): Candidate evidence - requires Quality review=1, Controlled current verification=11, Historical controlled evidence - stale or prior version=1, Firmware traceability signal - informational, not controlled verification evidence=2. CAPA-blocking evidence-class item count: 4. Rule: Risk evidence confidence is the average of per-item evidence-class weights (controlled_verification=1.00, historical_controlled=0.70, simulated=0.45, candidate=0.25, no_evidence=0.00). The final label is capped below high when any CAPA-closure-blocking evidence class is present, because those items cannot support final closure until controlled verification is attached.

Insufficient Evidence Gate

This CAPA remains open for insufficient evidence. Closure blocked: evidence class requires controlled verification before closure: RUN-TC-MEAS-032-V3-4-2, CAPA-CPL-100-013. Attach current controlled verification evidence, resolve synthetic/candidate/historical evidence gaps, and obtain Quality/Regulatory approval before closure.

Problem Statement

A low priority complaint was opened for DocPlus+ Pulse Oximeter. The user reported: Device brightness cant be adjusted. In practical terms, the concern is that the device may not perform as expected and the user may need additional support or alternate monitoring. The complaint record confirms the reported condition; the exact technical cause and final correction remain under investigation until objective evidence is reviewed.

Evidence class gate: Candidate evidence - requires Quality review, Controlled current verification, Historical controlled evidence - stale or prior version, Firmware traceability signal - informational, not controlled verification evidence

Complaint Summary

Reported condition: Device brightness cant be adjusted. Affected area: System Reliability. Reported timing or trigger: Not fully specified by complaint text. Observed symptom signals: brightness, cant, adjusted. No patient or user injury is stated in the complaint text; the risk review must still confirm potential safety impact.

Evidence class gate: Candidate evidence - requires Quality review, Controlled current verification, Historical controlled evidence - stale or prior version, Firmware traceability signal - informational, not controlled verification evidence

Immediate Containment Actions

Preserve the complaint evidence for System Reliability, including returned-unit information, event logs, service records, photos, user statements, configuration, lot or serial traceability, and software or hardware version where applicable. Screen recent complaints for similar symptoms and issue interim support, service, or use instructions if the risk review shows potential user or patient impact.

Evidence class gate: Candidate evidence - requires Quality review, Controlled current verification, Historical controlled evidence - stale or prior version, Firmware traceability signal - informational, not controlled verification evidence

Investigation

Relevant device requirements, verification records, risk controls, and open quality actions were reviewed for the complaint scope. The investigation focused on the reported condition and the affected area: System Reliability. For this complaint the scoped area is System Reliability, with emphasis on returned-unit evidence, complaint history, applicable requirements, risk controls, and verification records. Engineering and Quality reviewed the available design expectations, test records, risk controls, event evidence, and related quality actions. 1 verification record needs Quality review. The scoped verification evidence is available for Quality review. Risk evidence basis: Weighted evidence-class score 0.86 from 15 item(s): Candidate evidence - requires Quality review=1, Controlled current verification=11, Historical controlled evidence - stale or prior version=1, Firmware traceability signal - informational, not controlled verification evidence=2. CAPA-blocking evidence-class item count: 4.

Evidence class gate: Candidate evidence - requires Quality review, Controlled current verification, Historical controlled evidence - stale or prior version, Firmware traceability signal - informational, not controlled verification evidence

Root Cause Analysis

The current root-cause hypothesis is: A configuration error in the software may cause the brightness adjustment feature to be unresponsive. This is a probable cause, not a final confirmed cause, because the CAPA record still needs returned-unit review, complaint evidence, trend review, and approved verification evidence showing whether the suspected cause is truly responsible.

Evidence class gate: Candidate evidence - requires Quality review, Controlled current verification, Historical controlled evidence - stale or prior version, Firmware traceability signal - informational, not controlled verification evidence

Why CAPA Remains Open

The CAPA status is open because the available information is enough to justify corrective action, but not enough to close the quality record. Closure requires missing decisions or evidence: confirm or rule out the root cause, complete the correction or documented non-reproducibility rationale, verify the affected function under the reported use condition with approved testing, and obtain Quality/Regulatory approval of the residual risk decision. Risk evidence confidence is medium (score 0.86); Medium uncertainty: 11 of 15 evidence item(s) are current controlled verification, but 4 item(s) remain historical, simulated, candidate, or no-evidence class and cannot support CAPA closure. This CAPA remains open because the scoped evidence chain is insufficient for closure; Closure blocked: evidence class requires controlled verification before closure: RUN-TC-MEAS-032-V3-4-2, CAPA-CPL-100-013.

Evidence class gate: Candidate evidence - requires Quality review, Controlled current verification, Historical controlled evidence - stale or prior version, Firmware traceability signal - informational, not controlled verification evidence

Risk Assessment

The main hazard is that the device may not perform as expected and the user may need additional support or alternate monitoring. The current risk level is Medium and the complaint should remain in active risk review under ISO 13485 until patient impact, recurrence potential, and residual risk acceptability are documented. Risk evidence confidence is medium (score 0.86); Medium uncertainty: 11 of 15 evidence item(s) are current controlled verification, but 4 item(s) remain historical, simulated, candidate, or no-evidence class and cannot support CAPA closure.

Evidence class gate: Candidate evidence - requires Quality review, Controlled current verification, Historical controlled evidence - stale or prior version, Firmware traceability signal - informational, not controlled verification evidence

Corrective Actions

Reproduce or simulate the reported condition for System Reliability under the reported use conditions. Engineering and Quality should review returned-unit evidence, complaint history, applicable requirements, risk controls, and verification records, identify the confirmed failure mode, correct the issue or document why it cannot be reproduced. Rerun targeted verification, attach approved objective evidence, and resolve evidence gaps before release or closure.

Evidence class gate: Candidate evidence - requires Quality review, Controlled current verification, Historical controlled evidence - stale or prior version, Firmware traceability signal - informational, not controlled verification evidence

Preventive Actions

Update the design, service, or release checklist so future changes affecting System Reliability include complaint-specific verification for brightness, cant, adjusted. Add regression or trend monitoring where appropriate, and require evidence review before Quality approves a similar change or field disposition.

Evidence class gate: Candidate evidence - requires Quality review, Controlled current verification, Historical controlled evidence - stale or prior version, Firmware traceability signal - informational, not controlled verification evidence

Verification of Effectiveness

Effectiveness must be verified by successful targeted testing of the affected function under the reported use condition, review of complaint or service trends, and independent Quality review confirming that the affected area (System Reliability) meets approved design and risk-control requirements. The effectiveness package must replace candidate, historical, synthetic, or no-evidence gaps with approved objective evidence.

Evidence class gate: Candidate evidence - requires Quality review, Controlled current verification, Historical controlled evidence - stale or prior version, Firmware traceability signal - informational, not controlled verification evidence

CAPA Effectiveness Criteria

The CAPA is effective when the reported condition no longer occurs or is formally ruled out, the behavior for the affected function under the reported use condition is verified against the approved specification, all targeted verification activities pass, sufficient objective evidence is attached, and no related complaint recurs during six months after the corrected release or field correction.

Evidence class gate: Candidate evidence - requires Quality review, Controlled current verification, Historical controlled evidence - stale or prior version, Firmware traceability signal - informational, not controlled verification evidence

Lessons Learned

Future complaint handling for System Reliability should connect user-reported symptoms, objective evidence, risk controls, and verification records early. Similar changes or service decisions should include test coverage for the reported condition and review of whether existing controls remain effective.

Evidence class gate: Candidate evidence - requires Quality review, Controlled current verification, Historical controlled evidence - stale or prior version, Firmware traceability signal - informational, not controlled verification evidence

CAPA Closure

Closure decision: keep the CAPA open for insufficient evidence. The issue may warrant corrective action, but the scoped evidence chain still contains evidence that cannot support closure. Attach current controlled verification evidence, resolve synthetic/candidate/historical evidence gaps, and obtain Quality/Regulatory approval before closure.

Evidence class gate: Candidate evidence - requires Quality review, Controlled current verification, Historical controlled evidence - stale or prior version, Firmware traceability signal - informational, not controlled verification evidence

Technical evidence identifiers, requirement IDs, audit findings, and source citations are retained in the evidence and audit appendices. They are not repeated here so the CAPA remains readable for Quality, Engineering, Regulatory, and non-technical reviewers.

11.1 CAPA Action Plan

CAPA Record	CAPA-20260708-4268
Complaint Reference	CPL-20260708-3894
CAPA Type	Corrective and preventive action
Current Status	Pending - Insufficient Evidence
Insufficient Evidence Gate	This CAPA remains open for insufficient evidence. Closure blocked: evidence class requires controlled verification before closure: RUN-TC-MEAS-032-V3-4-2, CAPA-CPL-100-013. Attach current controlled verification evidence, resolve synthetic/candidate/historical evidence gaps, and obtain Quality/Regulatory approval before closure.

Action Plan

Purpose	Owner	Required Action	How Closure Is Proven
Immediate measures	Quality / Support	Preserve the complaint evidence for System Reliability, including returned-unit information, event logs, service records, photos, user statements, configuration, lot or serial traceability, and software or hardware version where applicable. Screen recent complaints for similar symptoms and issue interim support, service, or use instructions if the risk review shows potential user or patient impact.	Evidence is preserved, affected users are protected, and deployment risk is understood.
Fix the issue	Engineering	Reproduce or simulate the reported condition for System Reliability under the reported use conditions. Engineering and Quality should review returned-unit evidence, complaint history, applicable requirements, risk controls, and verification records, identify the confirmed failure mode, correct the issue or document why it cannot be reproduced. Rerun targeted verification, attach approved objective evidence, and resolve evidence gaps before release or closure.	The confirmed cause is fixed or ruled out with approved objective evidence.
Prevent recurrence	Engineering / Quality	Update the design, service, or release checklist so future changes affecting System Reliability include complaint-specific verification for brightness, cant, adjusted. Add regression or trend monitoring where appropriate, and require evidence review before Quality approves a similar change or field disposition.	Future changes or service decisions include controls for this failure mode before approval.
Check effectiveness	Quality Assurance	Effectiveness must be verified by successful targeted testing of the affected function under the reported use condition, review of complaint or service trends, and independent Quality review confirming that the affected area (System Reliability) meets approved design and risk-control requirements. The effectiveness package must replace candidate, historical, synthetic, or no-evidence gaps with approved objective evidence.	No repeat issue is seen after targeted verification and evidence gaps are closed.

Acceptance Criteria for Closure

Closure Gate	Required Evidence
Reported condition	The reported condition is reproduced and corrected, or formally ruled out with approved evidence.
Affected function	The affected function under the reported condition is verified against the approved requirement or acceptance criterion.
Verification	Missing, candidate, historical, synthetic, or no-evidence items are replaced by approved objective evidence.
Complaint monitoring	No related recurrence is observed during the defined follow-up period.
Approval	Quality and Regulatory reviewers approve the completed CAPA record.

12. Cybersecurity / SBOM Narrative

This section is device-level cybersecurity surveillance. It is not complaint evidence and it does not change the Risk Agent score automatically. It records what the Cybersecurity Agent found by reading the SBOM and querying the National Vulnerability Database for the CPE candidates in the workbook.

SBOM components	35
Components queried	25
NVD findings	7
Query date	2026-07-04
Source type	extracted
Cache status	cache_hit

Executive Severity Rollup

Critical	High	Medium	Low	None
0	2	5	0	0

VEX-style exploitability status is shown per CVE. The current automated SBOM/NVD match does not include device-specific exploitability determinations, so findings default to Under Investigation until Engineering documents whether the vulnerable code path is affected, fixed, not affected, or not reachable in this device configuration.

SBOM Component Coverage

Component	Version	Supplier	CPE Candidate	NVD Result
Bootloader	1.8.0	Device OEM / Secure Boot Team	cpe:2.3:o:device_oem:pulseox_bootloader:1.8.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Embedded Firmware	2.4.3	Device OEM / Embedded Software Team	cpe:2.3:o:device_oem:pulseox_firmware:2.4.3:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Operating System / RTOS	10.5.1	Amazon Web Services, Inc.	cpe:2.3:o:amazon:freertos:10.5.1:****:*	open_cves_found: 1 CVE record(s) returned by NVD for this CPE.
Device Drivers	1.6.2	Device OEM / Driver Integration Team	cpe:2.3:a:device_oem:pulseox_device_drivers:1.6.2:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Signal Processing Software	3.2.1	Device OEM / Clinical Algorithm Supplier	cpe:2.3:a:device_oem:pulseox_signal_processing:3.2.1:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Measurement Control Logic	2.1.0	Device OEM / Embedded Software Team	cpe:2.3:a:device_oem:pulseox_measurement_control:2.1.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Alarm Management Module	2.0.5	Device OEM / Safety Software Team	cpe:2.3:a:device_oem:pulseox_alarm_manager:2.0.5:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
User Interface Module	1.7.4	Device OEM / UI Software Team	cpe:2.3:a:device_oem:pulseox_ui_module:1.7.4:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Settings Configuration Manager	1.9.1	Device OEM / Embedded Software Team	cpe:2.3:a:device_oem:pulseox_settings_manager:1.9.1:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Data Storage Manager	1.5.0	Device OEM / Data Platform Team	cpe:2.3:a:device_oem:pulseox_data_storage_manager:1.5.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Patient Record Manager	1.3.2	Device OEM / Data Platform Team	cpe:2.3:a:device_oem:pulseox_patient_record_manager:1.3.2:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.

Component	Version	Supplier	CPE Candidate	NVD Result
Communication Stack	2.2.0	Device OEM / Connectivity Team	cpe:2.3:a:device_oem:pulseox_communication_stack:2.2.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Bluetooth Stack	5.3.0	Nordic Semiconductor ASA	cpe:2.3:a:nordicsemi:bluetooth_low_energy_stack:5.3.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Wi-Fi Stack	2.10	w1.fi project / Device OEM Integration	cpe:2.3:a:w1.fi:wpa_supplicant:2.10:****:	open_cves_found: 1 CVE record(s) returned by NVD for this CPE.
TCP/IP Stack	2.1.3	lwIP project	cpe:2.3:a:lwip_project:lwip:2.1.3:****:***	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
USB Interface Software	1.4.1	Device OEM / Connectivity Team	cpe:2.3:a:device_oem:pulseox_usb_interface:1.4.1:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
File System	3.8.1	littlefs project / Device OEM Integration	cpe:2.3:a:littlefs_project:littlefs:3.8.1:****:***	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Logging Module	1.2.8	Device OEM / Platform Software Team	cpe:2.3:a:device_oem:pulseox_logging_module:1.2.8:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Error Handling Module	1.1.6	Device OEM / Platform Software Team	cpe:2.3:a:device_oem:pulseox_error_handling:1.1.6:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Time Management Module	1.0.9	Device OEM / Platform Software Team	cpe:2.3:a:device_oem:pulseox_time_manager:1.0.9:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Self-Test Module	1.3.5	Device OEM / Test Software Team	cpe:2.3:a:device_oem:pulseox_self_test:1.3.5:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Firmware Update Manager	2.0.2	Device OEM / Release Engineering Team	cpe:2.3:a:device_oem:pulseox_update_manager:2.0.2:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Authentication Module	1.2.0	Device OEM / Security Software Team	cpe:2.3:a:device_oem:pulseox_authentication_module:1.2.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.

Component	Version	Supplier	CPE Candidate	NVD Result
Authorization Module	1.2.0	Device OEM / Security Software Team	cpe:2.3:a:device_oem:pulseox_authorization_module:1.2.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Cryptographic Library	3.0.13	OpenSSL Software Foundation	cpe:2.3:a:openssl:openssl:3.0.13:****:*	open_cves_found: 5 CVE record(s) returned by NVD for this CPE.

NVD CVE Findings - Structured Vulnerability Records

Vulnerability Record - CVE-2024-28115

CVE ID / NVD Reference	CVE-2024-28115 https://nvd.nist.gov/vuln/detail/CVE-2024-28115
CVSS Score / Severity / Vector	HIGH / CVSS 8.8 / v3.1 Vector: CVSS:3.1/ AV:L/ AC:L/ PR:L/ UI:N/ S:C/ C:H/ I:H/ A:H
CWE ID(s)	CWE-284: Improper Access Control, NVD-CWE-Other: Other weakness classification
Affected Component + Version	Operating System / RTOS 10.5.1
Vulnerability Description	NVD classifies the CVE under CWE-284, NVD-CWE-Other; review the public description for deployment-specific impact.
Exploitability / Exposure Context	Local or physically mediated exploitation should be assessed against service, update, or peripheral workflows. SBOM exposure: No direct exposure unless network stacks or external interfaces run as RTOS tasks.
Recommended Controls	review vendor fix status, confirm exploit preconditions, and document whether compensating controls cover the affected interface. SBOM controls: Minimal RTOS configuration, stack overflow checks, watchdog, task priority review, patch monitoring.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2024-28115

Vulnerability Record - CVE-2023-52160

CVE ID / NVD Reference	CVE-2023-52160 https://nvd.nist.gov/vuln/detail/CVE-2023-52160
CVSS Score / Severity / Vector	MEDIUM / CVSS 6.5 / v3.1 Vector: CVSS:3.1/ AV:N/ AC:L/ PR:N/ UI:R/ S:U/ C:H/ I:N/ A:N
CWE ID(s)	CWE-287: Improper Authentication
Affected Component + Version	Wi-Fi Stack 2.10
Vulnerability Description	NVD describes an authentication or access-control bypass condition in the affected component.
Exploitability / Exposure Context	Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Wi-Fi radio, WPA2/WPA3 negotiation, outbound cloud network.
Recommended Controls	update the TLS library, enforce strict certificate validation, and regression-test invalid certificate cases. SBOM controls: WPA2/WPA3, credential encryption, no ad-hoc mode, network allow-list, update monitoring.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2023-52160

Vulnerability Record - CVE-2009-1390

CVE ID / NVD Reference	CVE-2009-1390 https://nvd.nist.gov/vuln/detail/CVE-2009-1390
CVSS Score / Severity / Vector	MEDIUM / CVSS 6.8 / v2.0 Vector: AV:N/ AC:M/ Au:N/ C:P/ I:P/ A:P
CWE ID(s)	CWE-287: Improper Authentication
Affected Component + Version	Cryptographic Library 3.0.13
Vulnerability Description	NVD describes incomplete TLS certificate-chain validation, where a connection may be accepted because one certificate in an otherwise untrusted chain is trusted.
Exploitability / Exposure Context	Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Used by outbound TLS, update signature verification, and local encryption features.
Recommended Controls	verify full certificate-chain validation, require trusted root and hostname match, and regression-test rejected-chain cases. SBOM controls: Disable weak ciphers, certificate pinning, patch monitoring, crypto configuration review.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2009-1390

Vulnerability Record - CVE-2009-3765

CVE ID / NVD Reference	CVE-2009-3765 https://nvd.nist.gov/vuln/detail/CVE-2009-3765
CVSS Score / Severity / Vector	MEDIUM / CVSS 6.8 / v2.0 Vector: AV:N/ AC:M/ Au:N/ C:P/ I:P/ A:P
CWE ID(s)	CWE-310: Cryptographic Issues
Affected Component + Version	Cryptographic Library 3.0.13
Vulnerability Description	NVD describes certificate host-name validation that mishandles embedded NUL characters in the certificate Common Name, enabling a crafted certificate-name mismatch.
Exploitability / Exposure Context	Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Used by outbound TLS, update signature verification, and local encryption features.
Recommended Controls	reject embedded NUL characters in certificate names, prefer patched TLS hostname-validation routines, and test crafted certificate names. SBOM controls: Disable weak ciphers, certificate pinning, patch monitoring, crypto configuration review.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2009-3765

Vulnerability Record - CVE-2009-3766

CVE ID / NVD Reference	CVE-2009-3766 https://nvd.nist.gov/vuln/detail/CVE-2009-3766
CVSS Score / Severity / Vector	MEDIUM / CVSS 6.8 / v2.0 Vector: AV:N/ AC:M/ Au:N/ C:P/ I:P/ A:P
CWE ID(s)	CWE-310: Cryptographic Issues
Affected Component + Version	Cryptographic Library 3.0.13
Vulnerability Description	NVD describes a certificate or TLS validation weakness in the affected secure-communication path.
Exploitability / Exposure Context	Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Used by outbound TLS, update signature verification, and local encryption features.
Recommended Controls	update the TLS library, enforce strict certificate validation, and regression-test invalid certificate cases. SBOM controls: Disable weak ciphers, certificate pinning, patch monitoring, crypto configuration review.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2009-3766

Vulnerability Record - CVE-2009-3767

CVE ID / NVD Reference	CVE-2009-3767 https://nvd.nist.gov/vuln/detail/CVE-2009-3767
CVSS Score / Severity / Vector	MEDIUM / CVSS 4.3 / v2.0 Vector: AV:N/ AC:M/ Au:N/ C:N/ I:P/ A:N
CWE ID(s)	CWE-295: Improper Certificate Validation
Affected Component + Version	Cryptographic Library 3.0.13
Vulnerability Description	NVD describes certificate host-name validation that mishandles embedded NUL characters in the certificate Common Name, enabling a crafted certificate-name mismatch.
Exploitability / Exposure Context	Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Used by outbound TLS, update signature verification, and local encryption features.
Recommended Controls	reject embedded NUL characters in certificate names, prefer patched TLS hostname-validation routines, and test crafted certificate names. SBOM controls: Disable weak ciphers, certificate pinning, patch monitoring, crypto configuration review.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2009-3767

Vulnerability Record - CVE-2019-0190

CVE ID / NVD Reference	CVE-2019-0190 https://nvd.nist.gov/vuln/detail/CVE-2019-0190
CVSS Score / Severity / Vector	HIGH / CVSS 7.5 / v3.1 Vector: CVSS:3.1/ AV:N/ AC:L/ PR:N/ UI:N/ S:U/ C:N/ I:N/ A:H
CWE ID(s)	NVD-CWE-noinfo: No CWE information
Affected Component + Version	Cryptographic Library 3.0.13
Vulnerability Description	NVD describes a denial-of-service condition that may let an attacker disrupt availability of the affected software path.
Exploitability / Exposure Context	Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Used by outbound TLS, update signature verification, and local encryption features.
Recommended Controls	patch affected code paths, constrain exposed interfaces, and confirm watchdog/recovery behavior for availability loss. SBOM controls: Disable weak ciphers, certificate pinning, patch monitoring, crypto configuration review.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2019-0190

13. Electronic Records - 21 CFR Part 11

Requirement	DocPlus+ Implementation	Part 11
System Validation	DocPlus+ generation path is deterministic and testable through FastAPI route and PDF render checks.	11.10(a)
Audit Trail	Generation record links CPL-20260708-3894, CAPA-20260708-4268, timestamp, and agent trace events.	11.10(e)
Access Controls	AI output is marked draft and requires QE/RA approval before controlled use.	11.10(d)
Electronic Signatures	Signature block captures reviewer, approver, date, and meaning of signature.	11.50
Record Integrity	PDF is generated as a stable artifact with a deterministic document number and retrievable filename.	11.10(c)
Human-Readable Copy	Selectable-text PDF is available through the document download endpoint.	11.10(b)
Retention	Artifact is retained under output/pdf and can be indexed by M5 or a QMS repository.	11.10(c)

Generation Log Summary

Generation Hash	38944268
Input Fingerprint	38944268D373C928
AI Draft Status	Pending human quality and regulatory review
Low Confidence Handling	Evidence below 70% must be reviewed before release; current generated evidence remains visible with confidence values.

14. AuditShadow and Trace Decay

AuditShadow findings

Finding(s)	Requirement(s)	Risk	Observation	Remediation
FINDING-014	REQ-MEAS-032	Critical	Linked test evidence was executed on v3.4.2, but the device current firmware is v3.4.	Repeat verification on the current firmware or justify equivalence with approved rationale.

Trace Decay alerts

Requirement(s)	Test(s)	Tested FW	Required FW	Reason
REQ-MEAS-032	TC-MEAS-032	v3.4.2	v3.4	Test was executed on v3.4.2, while complaint firmware is v3.4.

14.1 Audit Data Interpretation and Corrective Action

AuditShadow and Trace Decay use the live M1 traceability graph as the controlled source for verification status. When the report flags missing evidence, it means the requirement does not have a formal VERIFIED_BY test link in that graph. It does not mean M4 found no supporting material. In this run, M4 retrieved 15 candidate evidence item(s), but candidate snippets remain draft support until a Quality reviewer links them to a requirement, test record, firmware version, and controlled source identifier.

Signal	What It Means	Source	Corrective Action
AuditShadow graph evidence gap	0 requirement finding(s) lack a VERIFIED_BY test link in M1.	M1 readiness score and traceability matrix	Create or link verification test evidence to each requirement and rescore readiness.
M4 retrieved evidence	15 candidate item(s) exist, but are not automatically controlled objective evidence.	M2 complaint pipeline and M4 evidence retrieval	Promote accepted snippets into controlled evidence nodes with requirement IDs, source, confidence, and firmware.
Trace Decay	1 firmware freshness alert(s) require review before release or approval.	M2 Trace Decay over M1 firmware/test links	Rerun verification on current firmware or document an approved equivalence rationale.
CAPA linkage	Open CAPA or complaint risk can keep readiness from being clean even when evidence exists.	M1 graph relationships and M2 risk assessment	Close, justify, or explicitly reference CAPA impact before final regulatory approval.

Reason and Corrective Action

The root issue is a traceability-linkage gap rather than a simple absence of all evidence. Corrective action is to synchronize M4 retrieved evidence back into the M1 graph as controlled verification evidence, then rerun AuditShadow and Trace Decay. Until that link exists, the report correctly treats the material as candidate support and keeps the audit finding open.

14.2 AuditShadow Detail Appendix

Finding(s)	Requirement(s)	Reference	Risk	Observation	Remediation
FINDING-014	REQ-MEAS-032	ISO 14971	Critical	Linked test evidence was executed on v3.4.2, but the device current firmware is v3.4.	Repeat verification on the current firmware or justify equivalence with approved rationale.

Audit Interpretation

Major and critical findings should be triaged before report approval. A graph-linked evidence gap means the requirement is present in the M1 graph while still lacking a controlled verification test relationship acceptable for submission or inspection. Candidate RAG evidence must be reviewed, accepted, and linked before the finding can close.

14.3 Trace Decay Detail Appendix

Requirement(s)	Test(s)	Tested Firmware	Required Firmware	Reason
REQ-MEAS-032	TC-MEAS-032	v3.4.2	v3.4	Test was executed on v3.4.2, while complaint firmware is v3.4.

Traceability Impact Statement

Trace Decay identifies verification evidence that may no longer represent the active or proposed firmware state. These alerts are not final nonconformities by themselves, but they are mandatory review inputs for release, CAPA, and audit-readiness decisions.

15. Cross-Standard Rendering Matrix

Element	FDA / QMSR	EU MDR	ISO 13485
Safety Requirement	Design input under QMSR; linked to verification evidence.	GSPR and Annex I safety/performance claim.	ISO 13485 Clause 7.3.3 design input.
Test Evidence	DHF verification record; current configuration/version required.	Annex II Section 6.1 V&V evidence.	Clause 7.3.6 design verification.
Complaint Record	Complaint file CPL-20260708-3894 with malfunction assessment.	Post-market vigilance signal under Articles 83/87.	Complaint handling under Clause 8.2.2.
CAPA Record	CAPA-20260708-4268 opened for System Reliability.	Trend/CAPA input for PMS and vigilance review.	Corrective action under Clause 8.5.2.
Risk Documentation	Risk analysis tied to design controls and complaint file.	Benefit-risk update in Annex II Section 5.	Risk management continuity through production/post-production.
Product Documentation	Lifecycle impact for System Reliability.	Design information and change-control evidence.	Validation and change-control evidence required.

Regulatory Lens Note

The same live graph and agent state are rendered under three regulatory lenses. This report therefore does not duplicate content blindly; it reformats the same evidence chain for FDA/QMSR, EU MDR, and ISO 13485 review contexts.

16. Document Control and Approval

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Complaint ID	CPL-20260708-3894
CAPA ID	CAPA-20260708-4268
Prepared by	DocPlus+ M4 Evidence and Documentation Layer
Generated from	Live M2 complaint pipeline, M1 graph context, M4 evidence retrieval, and device-level SBOM/NVD scan when attached
Generated at	2026-07-08T07:42:52Z
Agent status	completed
Classification	Confidential - AI Generated Draft
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Role	Name	Signature	Date
Quality Reviewer			
Regulatory Reviewer			